Unit 1: Introduction to Biology

1.1 Definition of Biology

Biology is the scientific study of living organisms (plants, animals, and microorganisms) and their interactions with each other and the environment.

It deals with:

- Structure (how living things are built)
- Function (how they work)
- Growth and development
- Origin and evolution
- Distribution of organisms

1.2 Why Do We Study Biology?

We study biology because it helps us to:

- **Understand life processes** (breathing, digestion, reproduction)
- **Improve health** – helps in medicine and disease control
- **Enhance agriculture** – better crop and animal production
- **Protect the environment** – conservation of natural resources
- **Develop technology** – biotechnology and scientific innovation
- **Solve everyday problems** related to food, health, and environment

1.3 The Scientific Method

The scientific method is a systematic way scientists use to investigate problems.

Steps:

1. **Observation** – identifying a problem
2. **Hypothesis** – a possible explanation
3. **Experimentation** – testing the hypothesis
4. **Data collection** – recording results
5. **Analysis** – interpreting data
6. **Conclusion** – accepting or rejecting the hypothesis

Important points:

- Experiments must be **controlled and repeatable**
- Variables:
    - Independent variable (changed)

- Dependent variable (measured)
- Controlled variables (kept constant)

1.4 Tools of a Biologist

Biologists use different tools in laboratories and in the field.

1.4.1 Laboratory Tools of Biologists

Used inside the lab:

- **Microscope** – observes tiny objects
- **Test tubes** – hold small samples
- **Beaker** – mixing and heating liquids
- **Flask** – storing and heating liquids
- **Petri dish** – culturing microorganisms
- **Bunsen burner** – heating substances
- **Thermometer** – measuring temperature
- **Balance** – measuring mass
- **Pipette/Dropper** – transferring liquids
- **Ruler** – measuring length

1.4.2 Field Tools

Used outside the laboratory:

- **Hand lens** – magnifies small objects
- **Binoculars** – observing distant organisms
- **Camera** – recording observations
- **Sweep net** – catching insects
- **Quadrat** – studying plant distribution
- **Measuring tape** – measuring distance
- **Notebook** – recording data

1.5 The Light Microscope

A light microscope is an instrument that uses light and lenses to magnify small objects so they can be seen clearly.

Importance:

- Helps study cells and microorganisms
- Essential for biological research

1.5.1 Parts and Functions of the Light Microscope

Main Parts:

- **Eyepiece (ocular lens)** – where you look
- **Objective lenses** – magnify objects (low, medium, high power)
- **Stage** – holds the slide
- **Stage clips** – keep slide in place
- **Mirror/Light source** – provides light
- **Coarse adjustment knob** – rough focusing
- **Fine adjustment knob** – sharp focusing
- **Arm** – supports upper parts
- **Base** – supports the microscope

1.5.2 Handling and Using a Light Microscope

Handling:

- Carry with **two hands** (arm and base)
- Keep it clean and dry
- Do not touch lenses with fingers

Steps to Use:

1. Place slide on stage
2. Start with low power objective
3. Adjust light source
4. Use coarse adjustment to focus
5. Use fine adjustment for clear image
6. Switch to higher magnification if needed

1.6 General Laboratory Safety Rules

- Follow teacher's instructions carefully
- Wear protective clothing (lab coat, gloves)
- Do not eat or drink in the lab
- Never taste chemicals
- Handle equipment carefully
- Report accidents immediately
- Keep the lab clean and organized
- Wash hands after experiments